

- 6 A set of 12 coloured filament light bulbs is designed for use with a 24 V d.c. supply. The 12 light bulbs are connected in series, as shown in Fig. 6.1.

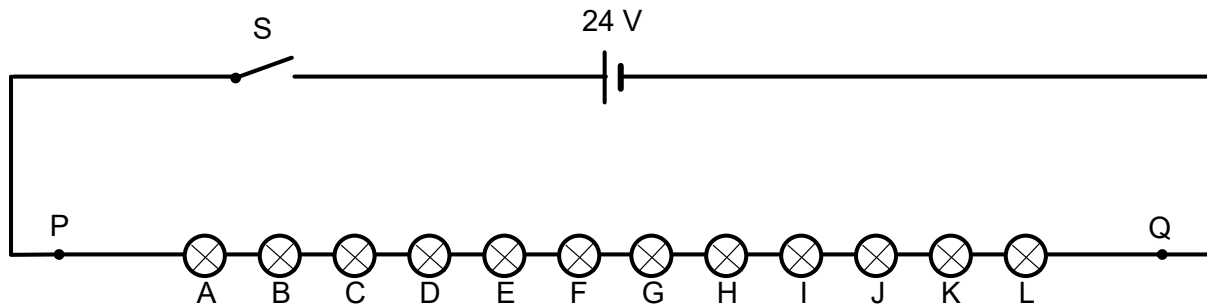


Fig. 6.1

The d.c. voltage supply has negligible internal resistance and each of the 12 light bulbs has the same constant electrical resistance.

- (a) When switch S is closed, the total power dissipated across the 12 light bulbs is 6.0 W.

Determine

- (i) the current passing through each light bulb,

current = A [1]

- (ii) the potential difference across each light bulb,

potential difference = V [1]

- (iii) the resistance of each light bulb

resistance = Ω [1]

- (b) In another situation, the light bulbs do not light up when switch S is closed. The fault is due to a single break in connection (open circuit) somewhere in the circuit.

A voltmeter is used to locate where this break in connection occurs.

Each of the following observations are **independent** of the other. State the possible location of the break in connection for each observation.

- (i) The potential difference between P and Q is zero.

.....
 [1]

- (ii) The potential difference is zero across every light bulb except H. The potential difference across H is 24 V.

.....
 [1]

- (iii) The potential difference between P and Q is 24 V but the potential difference is zero across every single light bulb.

.....
 [1]

- (c) A certain type of light bulb is designed such that the resistance of the bulb drops to zero when the filament fails. The light bulbs in Fig. 6.1 are replaced by this type of light bulb.

The filament of one light bulb fails when switch S is closed.

Determine

- (i) the potential difference across each of the remaining light bulb,

potential difference = V [1]

- (ii) the fractional increase in the power dissipated in each bulb.

fractional increase = [2]

[Total: 9]